

A NEW RENDERING AND PLUME SIMULATION FRAMEWORK FOR VISUALIZING TAAL VOLCANO ERUPTION PLUMES

Informed Consent Form

This consent form is written for computer science students and developers with a relevant background in 3D graphics or software development.

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Name of Research Project: A New Rendering and Plume Simulation Framework for Visualizing Taal Volcano Eruption Plumes

You may talk to anyone you feel comfortable talking with about the research, and you can take time to reflect on whether you want to participate. If you do not understand any of the words or concepts, we will take the time to explain them as we go, and you can ask questions at any time.

PURPOSE OF THE STUDY

You are being invited to take part in a research study. Before you decide to participate in this study, it is important that you understand why the research is being done and what your participation will involve. Please read the following information carefully, and feel free to ask the researcher if anything is unclear or if you need more information.

The purpose of the study is to evaluate a modernized 3D graphics rendering framework for an interactive Taal Volcano plume simulation (AnitoPlume V2). The study aims to assess whether the modernized system maintains an interactive and usable experience while evaluating rendering performance, graphics resource utilization, scalability, and visual fidelity. You are being invited to participate because you have a relevant background in computer science, software development, or 3D graphics, which is necessary to navigate and evaluate the simulator's technical features effectively.

STUDY PROCEDURES

If you choose to participate, you will be asked to complete the following procedures in chronological order:

- **Screening:** You will complete a short pre-screening form confirming that you meet the

inclusion criteria, such as being a collegiate CS student, graphics programmer, or software developer with prior exposure to real-time graphics or game development.

- **Orientation:** You will be given a brief, neutral walkthrough of the interactive Taal Volcano Plume simulator's controls and features.
- **Usability Task:** You will use the software to recreate specific Taal Volcano eruption plumes following a set of predefined scenarios.
- **Post-Session Questionnaire:** You will complete an anonymous, self-administered questionnaire using the System Usability Scale (SUS) to evaluate application responsiveness, stability, and visual parity.
- **Feedback:** You will participate in a closing interview to gather your suggestions for improving the interface and overall performance.

DURATION

The usability testing phase for the entire study will take place over 2 to 4 weeks. Your individual testing session will be kept brief to prevent eye strain and visual fatigue.

VOLUNTARY PARTICIPATION

Your participation in this study is voluntary. It is up to you whether or not you decide to participate. If you decide to participate, you will be asked to sign this consent form. After you sign this consent form, you are still free to withdraw at any time and without giving a reason. Withdrawing from this study will not affect the relationship you have, if any, with the researcher. You may exit the software at any stage and can choose to withdraw from the evaluation without academic or personal penalty. If you withdraw from the study before data collection is completed, your data will be destroyed.

RISKS

There are minimal risks associated with this study.

- **Physical:** You may experience minor eye strain or visual fatigue from interacting with the 3D graphics simulator.
- **Technical:** You may experience frustration if you encounter bugs or usability issues with the experimental prototype.

To mitigate these risks, researchers will provide immediate technical support for major difficulties. You are encouraged to request breaks at any time if you feel tired or frustrated, and the testing sessions will be kept brief.

You may decline to answer any or all questions and you may withdraw your participation at any time if you choose.

BENEFITS

There will be no direct financial benefit to your participation in the study. However, we hope that the information from this study will help establish a scalable rendering foundation that could potentially serve as one of the first accessible 3D visualization tools for volcanic hazard communication in the Philippines.

CONFIDENTIALITY

Your responses in this research will be anonymous. Every effort will be made by the researcher to preserve your confidentiality, including the following:

- All performance logs and usability questionnaire responses will be fully anonymized using alphanumeric identifiers.
- No personally identifiable information (PII) such as names, student numbers, or email addresses will be collected or tied to your responses.
- Survey sheets and digital spreadsheets will be secured in a password-protected cloud storage repository accessible only to the primary student researchers.
- Anonymized data will be kept for 4 to 5 years following final publication or degree conferral to allow for reproducibility audits before permanent deletion.
- Research results for publication will not include any personal identifiers of the research participant, unless explicit consent was given.
- Research participant will be notified and a separate consent will be asked should any collected data be used beyond the scope of this consent.

CONTACT INFORMATION

If you have any questions at any time about this study, or if you experience any non-normative sensations because of participation, or to exercise your data privacy rights, you may contact the researcher whose contact information is on the first page. If you have any questions regarding your rights as a research participant, or if problems arise which you do not feel you can discuss with the Principal Investigator, please contact the Chair of the DLSU Research Ethics Review Committee at chairerc@dlsu.edu.ph (632) 524-4611 local 513.

CONSENT

This section is mandatory I have read the provided information, or it has been read to me. I have had the opportunity to ask questions about it and any questions I have asked have been answered to my satisfaction. I understand that I will be given a copy of this form, and the researcher will keep another copy on file. I consent voluntarily to be a participant in this study.

Print Name of Participant _____

Signature of Participant _____

Date _____

Day/month/year

This section is mandatory

Print Name of Impartial Witness _____

Signature of Impartial Witness _____

Date _____

Day/month/year

This section is mandatory

Print Name of Researcher _____

Signature of Researcher _____

Date _____

Day/month/year

If the participant is illiterate¹

I have witnessed the accurate reading of the consent form to the potential participant, and the individual has had the opportunity to ask questions. I confirm that the individual has given consent freely.

Print name of witness _____

Signature of witness _____

Date _____

Day/month/year

¹ A literate witness must sign (if possible, this person should be selected by the participant and should have no connection to the research team).